Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	High productive inverse laser micro drilling with ultrashort pulse lasers of transparent materials for TGV production and material processing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery-Materials		Electrics & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	Optical energy-applied processing machines	Optical Application Devices	Laser Processing Machines
Concept of Proposed Technology	• Development of multibeam drilling process of large area substrates out of glass, sapphire or other transparent materials ○ Industry ready laser drilling process without the need of a subsequent etching process ○ Possibility to create different wall angles to adjust the hole taper ○ Realization of flexible hole patterns			
Rationale for Joint R&D Project	• Industrial companies using through glass vias necessary to understand demand of the industry ○ What is demand of geometry, quality, drilling speed? • Fraunhofer ILT only has know how in fabrication not usage • Metallization step and testing of electrical performance can not be done at Fraunhofer ILT			
Key Words	1. Through Glass Via laser processing			

	2. Laser drilling of transparent materials 3. Parallelization via multi beams

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (5-6)

R&D Objectives	• Development of inverse ultrashort pulse laser drilling process for small hole diameter
	  ○ Laser beam gets focused through transparent material on the backside or underside of thin substrate do ablate the material from the bottom up
	  ○ In contrast to SLE the material is directly ablated and not only modified and must be etched subsequently
	  ○ Challenge is to get ablated material out of the drilling hole
	• Investigation of minimum hole diameter and maximum aspect ratio for this process
	• Upscaling of inverse laser drilling process with multibeam optical systems for high productivity, optionally with flexible beam patterns			

R&D Target Specifications	• Development of Drilling process with 10-100 beams in parallel • Aspect ratio of bore holes at least 5 to 1 and up to 20 to 1 • Hole diameter 5-100 µm • Glass thickness 200-1000µm			

R&D Plan	Key tasks:
	• Development of inverse laser drilling process with ultra short pulse lasers
	• Upscaling of process from single beam to multi beam setup
	• Testing through glass vias performance with partner from korea
	
	Methodologies:
	• Laser process development for USP laser drilling includes variation of:
	  ○ Pulse duration (sub 100 fs to several ps)
	  ○ Pulse energy
	  ○ Bursts (MHz/GHz)
	  ○ Scan strategies
	• Upscaling of single beam process to multi beam optic using high power USP Laser and beam forming with DOE
	
	Timeline:
	• Approximated project duration: 36 months
	• Year 1:
	  ○ Development of single beam process for inverse laser drilling
	  ○ Milestone: expected target specification achieved with single beam
	• Year 2:
	  ○ Upscaling to 10-100 beams in parallel
	  ○ Milestone: parallelization with same process quality achieved
	• Year 3:
	  ○ Testing of different geometries, hole shapes with korean partner
	  ○ Metallization and testing done by Korean SME
	  ○ Milestone: Performance sufficient for Korean SME			

4. R&R (Roles and Responsibilities)

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Expected R&R of Research Institute	Role: Experts in laser-materials processing and optical technology • Scaling (multi beams, elongated beams, bursts, fast beam deflection) • High Precision (micro scanner und UV-laser systems) • High adaptability (Development and manufacturing of custom optical systems) Capabilities: • 420 employees with more than 11,000 m2 of usable floor space • >15 high precision USP laser machines (USP laser power up to 1 kW, wavelength from IR to DUV, pulse duration down to 50 fs, flexible beam shaping with >100 beamlets) • Well-equipped in-house metrology (SEM, AFM, LSM, WLI) Contributions • Development and building of large area laser drilling optics • Development of laser processes and parameters • Adapting of hole geometry based on partners needs • IP on ultrashort pulse-based precision laser ablation processes • IP on development and setup of optical processing heads for tailored intensity distributions or multi beam systems • Comprehensive collection of system technology for laser-based precision manufacturing • Access to ILT's supplier and industry partner network
Expected R&R of Korean SME Partner	Role: Experts in metallization and chip infrastructure • Defines target geometry and quality of laser drilling • Expected to contribute with metallization of through glass vias • testing of electrical performance and Quality control of machined part

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Laser drilling process for transparent materials such as glass or sapphire ○ without Etching step ○ Better Edge quality of entry and exit in comparison to selective laser etching. ○ High productivity • Invention or patents for this technology • Further public funded and bilateral projects for further industrialization of the technology towards TRL 9

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization